AOU
الجامعة العربية المفتوحة
Arab Open University
Faculty of Computer Studies

TM298
Operating Systems

Midterm Examination (MTA) - Makeup
Spring Semester 2025/2026

Date: 30/Apr./2026

Number of Exam Pages: 3 (including this cover sheet)	**Time Allowed:** 90 Minutes

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Instructions:

- Total Marks: **30**
- This exam consists of **three** parts.
- **ALL questions** must be answered in the External Answer booklet.
- Be sure you write your **name and ID** on the External Answer booklet.
- **Calculators** are allowed.

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Part 1: Multiple Choice Questions: [5 marks]

Question 1: Choose the most correct answer.

1. Multiprogramming improves system performance by:
   a) Increasing memory size.
   b) Reducing interrupts.
   c) Eliminating I/O.
   d) Increasing CPU utilization.

2. A trap is best described as:
   a) Device signal.
   b) Context switch.
   c) Software interrupt.
   d) Hardware failure

3. Ordinary pipes require:
   a) Network connection.
   b) Shared memory.
   c) Parent-child relationship.
   d) Kernel threads.

4. Which of the following is a type of system call?
   a) File manipulation
   b) Memory allocation
   c) Process control
   d) All the above

5. The one program always running on the computer is:
   a) Kernel
   b) Application programs
   c) System programs
   d) All the above

Part 2: Short Essay Questions: [15 marks]

Question 2: [2 marks]
Explain the main roles of operating systems.

Question 3: [2 marks]
Explain the purpose of the Process Control Block (PCB).

Question 4: [3 marks]
What do the following acronyms stand for?
- a- CLI
- b- IPC
- c- SAN

Question 5: [4 marks]
In a multithreading system, suppose in the user space there are only 4 threads in each model. Compare between **Many to One** and **Many to Many** multithreading models by drawing each model and showing how the four threads in the user space will be mapped to appropriate threads in the kernel space.

Question 6: [4 marks]
a) List the **different process states** used by an operating system and **explain what each state represents during a process's execution.** (2.5 marks)
b) Draw process state diagram. (1.5 marks)

Part 3: Problem Solving: [10 marks]

Question 7: [4 marks]
a) What is the meaning of the following crontab time format? (2 marks)

8 7-10 * * *

b) What does ***chmod*** do? Provide any valid example. (2 marks)

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Question 8: [6 marks]

Write a Bash script that prompts the user to enter **three grades** *using a loop, where:*

- The **first grade** represents the **TMA** (out of 20),
- The **second grade** represents the **MTA** (out of 30),
- The **third grade** represents the **Final Exam** (out of 50).

The script should then **calculate and display the final grade out of 100** *by summing the three* components according to their weights.

Notes:

- Assume all entered values are valid numbers *and within their correct ranges.*
- No validation is required.
- Do **not** use arrays or advanced Bash features.

Hint: Use if–elif-else statement *inside the loop to display the appropriate prompt for each grade.*

Expected Output (Example Run):

```
Enter TMA grade (out of 20): 16
Enter MTA grade (out of 30): 22
Enter Final Exam grade (out of 50): 40
The final grade is 78
```

End of Question

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